

L41 — Heap Sort and Comparative Analysis

Unit: 3 | Chapter: Sorting | BT Level: BT5 | CO: CO3, CO5

Learning Objectives

- Implement heap sort using a max-heap
 - Trace heap sort step by step
 - Prove $O(n \log n)$ time complexity
 - Compare heap sort against merge sort and quick sort
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1. Heap Sort Idea

Heap sort uses a max-heap to sort an array in-place:

1. **Build max-heap** from array — $O(n)$
2. **Repeatedly extract max:**
 - Swap root (max) with last element
 - Reduce heap size by 1
 - Sift down the new root
 - Repeat $n-1$ times — $O(n \log n)$

After $n-1$ extractions, the array is sorted in ascending order.

2. Implementation

```
def heap_sort(arr):  
    """  
    Sort array in-place using heap sort.  
    Time:  $O(n \log n)$  | Space:  $O(1)$   
    """  
    n = len(arr)  
  
    def sift_down(arr, root, size):  
        while True:  
            largest = root  
            left = 2 * root + 1  
            right = 2 * root + 2  
  
            if left < size and arr[left] > arr[largest]:  
                largest = left  
            if right < size and arr[right] > arr[largest]:  
                largest = right  
  
            if largest != root:  
                arr[root], arr[largest] = arr[largest], arr[root]
```

```

        root = largest
    else:
        break

# Step 1: Build max-heap
for i in range(n // 2 - 1, -1, -1):
    sift_down(arr, i, n)

# Step 2: Extract max n-1 times
for end in range(n - 1, 0, -1):
    arr[0], arr[end] = arr[end], arr[0] # Move max to sorted posi
    sift_down(arr, 0, end)             # Restore heap for reduce

return arr

```

3. Detailed Trace

Array: [4, 10, 3, 5, 1, 9]

Step 1: Build Max-Heap

Start from last internal node (index = $n//2 - 1 = 2$):

i	Sift down	Array
2	sift_down(2): children are 9 (idx 5). $9 > 3 \rightarrow$ swap	[4, 10, 9, 5, 1, 3]
1	sift_down(1): children 5(idx 3),1(idx 4). $10 > 5 \rightarrow$ no swap	[4, 10, 9, 5, 1, 3]
0	sift_down(0): children 10(idx1),9(idx2). $10 > 4 \rightarrow$ swap	[10, 4, 9, 5, 1, 3]
	Continue: 4 at idx 1, children 5(idx3),1(idx4). $5 > 4 \rightarrow$ swap	[10, 5, 9, 4, 1, 3]

Max-heap built: [10, 5, 9, 4, 1, 3]

Step 2: Extraction (heap_size shrinks each time)

Round	Swap root with	Array	Heap portion	Sorted portion
1	arr[0]↔arr[5]: 10↔3	[3, 5, 9, 4, 1, 10] → sift [3,5,9,4,1] → [9, 5, 3, 4, 1, 10]	[9, 5, 3, 4, 1]	[10]
2	arr[0]↔arr[4]: 9↔1	[1, 5, 3, 4, 9, 10] → sift → [5, 4, 3, 1, 9, 10]	[5, 4, 3, 1]	[9, 10]
3	arr[0]↔arr[3]: 5↔1	[1, 4, 3, 5, 9, 10] → sift → [4, 1, 3, 5, 9, 10]	[4, 1, 3]	[5, 9, 10]
4	arr[0]↔arr[2]: 4↔3	[3, 1, 4, 5, 9, 10] → sift → [3, 1, 4, 5, 9, 10]	[3, 1]	[4, 5, 9, 10]
5	arr[0]↔arr[1]: 3↔1	[1, 3, 4, 5, 9, 10]	done	all sorted

Sorted: [1, 3, 4, 5, 9, 10] ✓

4. Complexity Proof

Build heap: $O(n)$ (shown in L40)

n-1 extractions: Each extraction:

- Swap: $O(1)$
- Sift down: $O(\log k)$ where k = current heap size

Total extraction cost: $\sum_{k=2}^n \log k = \log(n!) = O(n \log n)$

By Stirling's approximation: $\log(n!) \approx n \log n - n \approx O(n \log n)$

Total: $O(n) + O(n \log n) = O(n \log n)$ for all cases.

5. Comparative Analysis

Property	Heap Sort	Merge Sort	Quick Sort	Insertion Sort
Best case	$O(n \log n)$	$O(n \log n)$	$O(n \log n)$	$O(n)$
Average case	$O(n \log n)$	$O(n \log n)$	$O(n \log n)$	$O(n^2)$
Worst case	$O(n \log n)$	$O(n \log n)$	$O(n^2)$	$O(n^2)$
Space	$O(1)$ in-place	$O(n)$	$O(\log n)$ avg	$O(1)$
Stable?	No	Yes	No (typical)	Yes
Cache friendly?	Poor	Good	Good	Excellent
Adaptive?	No	No	Partially	Yes
Practical speed	Slower (poor cache)	Fast	Fast	Fast for small n

6. Why Heap Sort is Slow in Practice

Despite $O(n \log n)$ worst-case, heap sort is often **slower than quicksort** in practice due to **poor cache locality**:

- Heap operations jump between distant array indices ($\text{child} = 2i + 1, 2i + 2$)
- These jumps cause frequent **cache misses** in modern CPUs
- Quick sort has better locality — it accesses contiguous memory regions
- Merge sort uses sequential access patterns (merge step)

Typical performance ranking (in practice):

Quick sort > Merge sort > Heap sort

When to use heap sort:

- When worst-case $O(n \log n)$ guarantee is required AND in-place sorting is needed
 - Embedded systems with limited memory ($O(1)$ space)
 - Real-time systems that need bounded running time
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7. k Largest / k Smallest Elements

```
import heapq

def k_largest(arr, k):
    """Find k largest elements efficiently using min-heap.  $O(n \log k)$ """
    return heapq.nlargest(k, arr)

def k_smallest(arr, k):
    """Find k smallest elements.  $O(n \log k)$ """
    return heapq.nsmallest(k, arr)

# Alternative: partial heap sort
def k_largest_manual(arr, k):
    """Build max-heap, extract max k times."""
    heap = arr[:]
    n = len(heap)
    # Build max-heap
    for i in range(n // 2 - 1, -1, -1):
        _sift_down(heap, i, n)

    result = []
    for _ in range(k):
        result.append(heap[0])
        heap[0] = heap[n - 1]
        n -= 1
        _sift_down(heap, 0, n)

    return result
```

Summary

- **Heap sort:** Build max-heap ($O(n)$), then repeatedly swap root with end and sift down ($O(n \log n)$).
 - Time: $O(n \log n)$ in **all cases** (best, average, worst) — guaranteed.
 - Space: $O(1)$ — in-place sorting.
 - Not stable (equal elements may be reordered).
 - Slower in practice than quick sort due to poor cache locality.
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References

- Cormen et al., *Introduction to Algorithms*, Ch. 6 (MIT Press, 2022)
- Skiena, *The Algorithm Design Manual*, Ch. 4.1 (Springer, 2020)